

Likelihood Functions

CSU, Chico Math 314

2025-09-15

outline

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Review: estimating parameters

Recall that statistics attempts to estimate the (population) parameters of interest via an assumed known functional form. The population parameters are in reality functions of unknown variables of a probability mass/distribution function.

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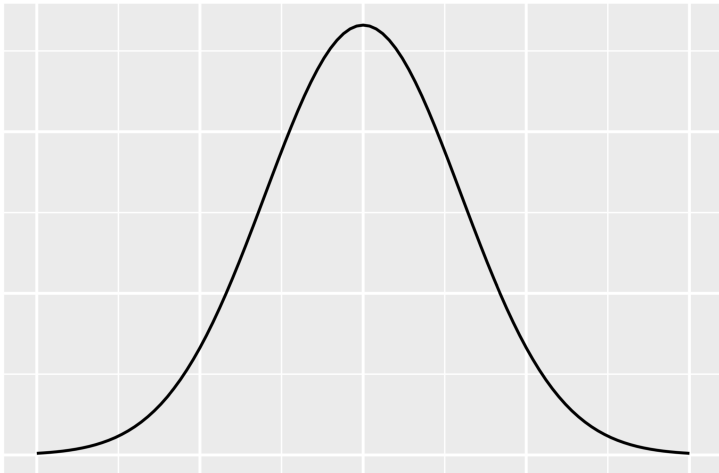
References

Motivating Likelihood

If we know the distribution function from which a sample of data was generated, we should be able to estimate the parameter(s) of the distribution that is most likely to have generated these data.

Motivating Likelihood

Suppose we randomly sampled 6 people and found heights 69, 72, 70, 68, 71, 71 inches. What are the parameters that are most likely to have brought about these data?



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Bernoulli Example

Suppose we observed $x_1, \dots, x_n \sim_{iid} \text{Bernoulli}(p)$. What is the maximum likelihood estimator for p ?

$$f(x|p) = p^x(1-p)^{1-x}$$

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Likelihood Function

Consider n observations $x_1, \dots, x_n$ independently drawn from the distribution $G(x|\theta)$. We seek to find the parameter θ that is most likely to have generated the data $\mathbf{x}$.

likelihood function

$$L(\theta|\mathbf{x}) = \prod_{i=1}^n G(x_i|\theta).$$

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Likelihood Function, notes

- ▶ We think about L as a function of θ instead of $\mathbf{x}$.
- ▶ All observations are independent and identically distributed, *iid*.
- ▶ $\hat{\theta} = \arg \max_{\theta} L$, is the mathematical equivalent of “parameters most likely to have generated the observed data.”
- ▶ It should be immediately clear that nobody wants to work with $L(\theta|\mathbf{x})$ itself.

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Maximum Likelihood

Estimators derived via the likelihood function are called **maximum likelihood estimators** or **MLEs**. MLE estimators estimate parameters of interest by maximizing the likelihood function. That is $\hat{\theta}$ is the MLE if

$$\hat{\theta} = \arg \max_{\theta} L(\theta|\mathbf{x}).$$

Maximum Likelihood, computation

So how does one find $\hat{\theta}$?

- ▶ Calculus.
- ▶ It's immensely helpful to work with the natural log of the likelihood function.
- ▶ Maximize L as a function of θ just as we would any other function f in calculus.

Detour the First

How can we rewrite (simplify?) the following equations?

- ▶ $\log(a \cdot b)$?
- ▶ $\log(x_1 \cdot x_2)$?
- ▶ $\log(x_1 \cdot x_2 \cdot \dots \cdot x_n)$?
- ▶ $\log(\prod_{i=1}^n x_i)$?
- ▶ $\log(\prod_{i=1}^n f(x_i))$?

Likelihood Function, computation

Use properties about the natural log to manipulate the likelihood function L into a reasonable function to optimize.

$$\ell(\theta|\mathbf{x}) = \log L(\theta|\mathbf{x}) = \sum_{i=1}^n \log G(x_i|\theta).$$

In other words, **the log of a product is a sum of logs.**

Log Likelihood Function, notes

The log-likelihood function ℓ has a two main advantages over L .

- ▶ ℓ is easier to optimize for both humans and computers.
- ▶ $\hat{\theta} = \arg \max_{\theta} \ell(\theta|\mathbf{x}) = \arg \max_{\theta} L(\theta|\mathbf{x})$ because $\log()$ is monotonically increasing.

In other words, both the likelihood function and the log-likelihood function have the **same maximum**.

Detour the Second

How can we rewrite (simplify?) the following equations?

- ▶ $\frac{d}{d\theta}\{a \cdot \theta + b \cdot \theta\}?$
- ▶ $\frac{d}{d\theta}\{x_1 \cdot \theta + x_2 \cdot \theta\}?$
- ▶ $\frac{d}{d\theta}\{x_1 \cdot \theta + \dots + x_n \cdot \theta\}?$
- ▶ $\frac{d}{d\theta}\{\sum_{i=1}^n x_i \cdot \theta\}?$

Detour the Third

Evaluate the following expression,

$$\frac{d}{d\theta} x \cdot \log(1 - \theta).$$

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Exponential Example

Suppose we observed $x_1, \dots, x_n \sim_{iid} \text{Exponential}(\lambda)$. What is the MLE for λ ?

$$f(x|\lambda) = \lambda \exp(-\lambda x)$$

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Note, we are going to brush over many topics that are often at least one class worth of material: optimization techniques.

```
?optim  
?rbinom  
?rpois
```


Using `optim`

The function `optim` is useful for finding MLEs, but it requires some setup to work effectively.

- ▶ First, solve for the log-likelihood function ℓ .
- ▶ This will require the same initial steps as working the function out manually.
- ▶ Then write ℓ in terms RStudio can understand by defining a *function* for ℓ
- ▶ Use this newly defined R function as an argument within the `optim` function.

Writing functions in RStudio

- ▶ You can define a function in Rstudio by using `function()`.
- ▶ Inside your function, define TWO arguments: a parameter θ and a data set x .
- ▶ Depending on the distribution in question, θ may consist of more than one value.

Example: A Function for Sample Mean

```
sample.mean <- function(x) sum(x)/length(x)  
sample.mean(1:10)
```

Function arguments

It is helpful to define some terms used for calculations within the function itself. This will cut down on the number of arguments required as input for your function. For example, rather than including the length n of the data set as an argument of the function, we can define n within the function itself, since we can obtain it so long as we have the data set x as an argument.

Arguments for `optim`

You can enter `?optim` in the Console to learn more about its arguments, but here are a few key points:

- ▶ `par`: Initial value(s) to test on the function. Note that this must be of equal length to the number of parameters.
- ▶ `fn`: The function to be minimized or maximized. This is where you put the name of your new function for the log likelihood.
- ▶ `method`: The optimization method used. "L-BFGS-B" will allow you to set lower and upper bounds, useful for distributions in a bounded range.
- ▶ `control`: `optim` defaults to minimizing the function. Set `control=list(fnscale=-1)` to maximize instead.
- ▶ Though it isn't an argument by default, you'll want to define the variable `x` within the arguments of `optim` to be equal to the data set so your function will accept it as an argument as well.

MLE of the Exponential Distribution using `optim`

Using the log likelihood function for the Exponential distribution from earlier this week, how can we use `optim` to solve for the MLE of λ ? Remember, the PDF is:

$$f(x|\lambda) = \lambda \exp(-\lambda x)$$

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references I

Michael G. Akritas. *Probability and Statistics with R for Engineers and Scientists*. Pearson Education, Inc., 2016.
Adapted from notes by Dr. Roualdes.